Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
Prof. Dr. Sofia C. Olhede

Practice Exam — Week 7: Spectral Density & Spectral Representation

Duration: 60 minutes · No calculator · Answer on paper

Name: ______________________ Surname: ______________________

SCIPER (Student Card Number): ______________________

Exercise	**Points**
1	
2	
3	
4	
Total	

Justify every answer. Throughout, ε_t is a mean-zero white noise of variance σ^2.

Problem 1 (5 points)

For a discrete-time stationary process $\{X_t\}_{t\in\mathbb{Z}}$ with summable ACVS $\{\gamma_\tau\} \in \ell^1$ (sampling interval $\Delta = 1$), the spectral density function (SDF) is $\mathcal{S}(f) = \sum_{\tau\in\mathbb{Z}} \gamma_\tau \, e^{-2\pi i f \tau}$, with inverse $\gamma_\tau = \int_{-1/2}^{1/2} \mathcal{S}(f) \, e^{2\pi i f \tau} \, df$.

(a) Prove from the definition that any SDF is real-valued and even, $\mathcal{S}(-f) = \mathcal{S}(f)$, and write it in the cosine form $\mathcal{S}(f) = \gamma_0 + 2\sum_{\tau\geq 1} \gamma_\tau \cos(2\pi f \tau)$. (1.5 pts)

(b) Show that $\gamma_0 = \mathrm{Var}(X_t) = \int_{-1/2}^{1/2} \mathcal{S}(f) \, df$, and use this to interpret $\mathcal{S}(f)$ as a distribution of power across frequency. (1.5 pts)

(c) Take $\{X_t\} = \{\varepsilon_t\}$ (white noise of variance σ^2). Compute $\mathcal{S}(f)$ directly from its ACVS and explain in one sentence why the result earns the name "white". (2 pts)

Problem 2 (5 points)

This problem tests whether a candidate sequence is a valid autocovariance.

(a) *(The validity criterion.)* State the criterion that decides whether a summable symmetric sequence $\{\gamma_\tau\}$ is the ACVS of *some* stationary process, expressed through its Fourier transform $\mathcal{S}$. Why is non-negativity of $\mathcal{S}$ the right test? (1 pt)

(b) *(A truncated ACVS.)* Let $\gamma_\tau = 1$ for $|\tau| \le K$ and $\gamma_\tau = 0$ for $|\tau| > K$, where K is a positive integer. Compute $\mathcal{S}(f)$ in closed form (a Dirichlet kernel), then evaluate it at the Nyquist endpoint $f = \frac{1}{2}$ to decide for which K the sequence *fails* to be a valid ACVS. (2 pts)

(c) *(MA(1)-shape ACVS.)* Let $\gamma_0 = 1$, $\gamma_{\pm 1} = \rho$, and $\gamma_\tau = 0$ for $|\tau| \ge 2$, with $\rho \in \mathbb{R}$. Compute $\mathcal{S}(f)$ and determine the exact set of ρ for which $\{\gamma_\tau\}$ is a valid ACVS. Relate the answer to the largest possible lag-1 autocorrelation of an MA(1). (2 pts)

Problem 3 (5 points)

Let $\{X_t\}$ be the MA(q) process $X_t = \sum_{j=0}^{q} \theta_j \varepsilon_{t-j}$ with $\theta_0 = 1$ and ε_t white noise of variance σ^2.

(a) Recall (or derive) the ACVS $\gamma_\tau = \sigma^2 \sum_{j=0}^{q} \theta_j \theta_{j+|\tau|}$ for $|\tau| \le q$ (and 0 otherwise), writing it in the double-sum form $\gamma_\tau = \sigma^2 \sum_{j=0}^{q} \sum_{k=0}^{q} \theta_j \theta_k \, \mathbf{1}_{\{\tau=k-j\}}$. (1.5 pts)

(b) Starting from this double sum, derive the factored SDF

$$\mathcal{S}(f) = \sigma^2 \left| \sum_{j=0}^{q} \theta_j e^{-2\pi i j f} \right|^2,$$

and explain in one sentence why this form makes $\mathcal{S}(f) \ge 0$ automatic. (2 pts)

(c) Specialise to the MA(1) $X_t = \varepsilon_t - \theta \varepsilon_{t-1}$. Give $\mathcal{S}(f) = \sigma^2\big(1 + \theta^2 - 2\theta \cos(2\pi f)\big)$, and for $\theta > 0$ locate the frequency of the maximum and of the minimum of $\mathcal{S}$, giving the two extreme values. (1.5 pts)

Problem 4 (5 points)

(Revision of Week 6: Fourier theory — convolution & aliasing.) Work with the continuous-time Fourier transform $G(f) = \int_{-\infty}^{\infty} g(t)e^{-2\pi i f t}\,dt$ and convolution $(g * h)(t) = \int_{-\infty}^{\infty} g(t-y)h(y)\,dy$ ($g, h \in L^1$).

(a) *(Convolution theorem.)* Prove that if $u = g * h$ then $U(f) = G(f)\,H(f)$, justifying carefully the interchange of integration (state the Fubini hypothesis you use). (2 pts)

(b) *(Aliasing.)* Let $g \in L^1$ be continuous with transform $\mathcal{G}$, and let G denote the transform of the sampled sequence $\{g(t)\}_{t\in\Delta\mathbb{Z}}$. Show the aliasing identity $G(f) = \sum_{k\in\mathbb{Z}} \mathcal{G}(f + k/\Delta)$, and define the Nyquist frequency. (2 pts)

(c) *(A concrete alias.)* A pure cosine of frequency $f_0 = 0.8$ Hz is sampled at rate $1/\Delta = 1$ Hz (so $\Delta = 1$ s). To which frequency in the principal Nyquist band $[-\frac{1}{2}, \frac{1}{2}]$ does f_0 fold, and what is the lowest sampling rate that would resolve f_0 without aliasing? (1 pt)